Marvin Wu(Wu Hung Mao)
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Research interest: reliable, efficient computer-vision systems on real-world hardware and data.

Education

Honours Bachelor of Science (HBSc), Computer Science specialist. Sep. 2021 - Dec. 2026
University of Toronto Mississauga (expected graduation with PEY CO-OP - December 2026)
GPA (as of Oct 2026): 3.44/4.0 cumulative | 3.54/4.0 last two years of study (excl. co-op)

Publications

Peer-reviewed publication Sep 2026
Khan, L., **Wu, H.-M.**, Lin, W., Bi, F., Abdelhadi, Y., Jung, J. "Ensemble Deep Learning Approaches for AI-Altered Video Detection." *Journal of Computing, Data, and Exploration (CODEX)*, vol. 2, no. 1, 2026. Preprint: arXiv:2607.06872.

- Presents a multimodal deepfake-detection ensemble (AASIST for audio; EfficientNet-B1, XceptionNet, and MesoNet for video frames) trained on AIGVDBench, FaceForensics++, and ASVspoof 2019 and evaluated on FakeAVCeleb, comparing five fusion strategies.

Course research manuscript Apr 2026
Riku, T., Ahmed, K., **Wu, H.-M.**, Usman, R. "Can Agents Debug Better Than a Software Engineer?" CSC398, University of Toronto Mississauga.

- Evaluates the bug-localization and automated-fix capabilities of LLM-based coding agents against the JetBrains LCA benchmark.

Research Experience

Gastric Cancer Detection from Endoscopy Videos Oct 2026 - Present
Department of Computer Science, University of Toronto *Supervisor: Dr. Ameya Pore, Postdoctoral Fellow*

- Joined a project with clinicians building an annotated upper-GI endoscopy video dataset and a real-time segmentation model for early-stage gastric cancer detection, with the goal of deploying it on the hospital's server.

Work Experience

Teaching Assistant — CSC105H5F (Artificial Intelligence Literacy) Sep 2026 - Dec 2026
University of Toronto Mississauga *Supervisor: Prof. Josh Jung*

- Hold weekly office hours and provide asynchronous student support via Piazza for an introductory AI literacy course of approximately 215 students.
- Prepare and teach 4-hour weekly lab sessions for a 40-student section, grade lab submissions, and invigilate examinations to support course assessment and academic integrity.

Physical Layer (vDU) Developer Intern May 2024 - Sep 2025
Ericsson *Ottawa, Ontario, Canada*

Project: Gphy2.0 (Feb 2025 - Sep 2025)

* Worked alongside senior engineers to move LDPC encoding from an Intel accelerator to NVIDIA GPUs, integrating the GPU branch into the existing codebase.
* Profiled latency of the GPU branch with Nsight Systems and Nsight Compute, comparing reports collected across servers with different GPUs (NVIDIA A10, A100, and Grace Hopper) and adjusting kernels to improve SM occupancy.
* Collected latency with the Intel accelerator branch calculated with the CPU cycle.
* Reported results to the team in daily sync-ups and worked with teammates to identify concurrency bugs.
* **Environment:** Bazel(`rules_cuda`), Gerrit, Nsight Systems, Nsight Compute

Project: Gemini (May 2024 - Feb 2025)

* Digital twin for Radio, RF Channel, and UEs.
* Created a constellation diagram using the Ericsson Design System library and Vue3 framework.
* Wrote multi-threaded unit tests in CUDA to test a real-time system under a strict timing constraint ($\sim$ 35 µs).
* Wrote an SQLite script that drew histograms based on data collected from LTTng traces for channel quality analysis.
* **Environment:** Docker, Wireshark, Gerrit, Gitlab, Jenkins, Jfrog, Sqlite3, AWS(EC2)

Computer Vision & Machine Learning Systems Projects

Fall Detection Capstone Project (CSC490) | *Hugging Face, Vision-Language Models, Fir HPC* Sep 2026 - Dec 2026

- Investigate whether specialized fall-detection models are more robust to camera-viewpoint changes than generic vision-language models, in a 4-person team using OmniFall, a multi-view dataset of 14,000+ real and synthetic fall videos (33+ hours, 16 labels).
- Set up GPU compute on the Fir HPC cluster (Digital Research Alliance of Canada, formerly Compute Canada) and downloaded the OmniFall dataset, which aggregates 10 relabelled source datasets.
- Analyze the dataset to scope the evaluation to the 5 sources that film the same fall from at least two camera angles; benchmark generic vision-language models (e.g., Kimi K3) for the generic-vs-specialized comparison.
- Evaluate under a controlled protocol (identical test set, resolution, preprocessing, and labels; no training on the test set), reporting accuracy, precision, recall, F1, specificity, and confusion matrices separately for each camera view.

AI-Video Detection Ensemble | *Python, PyTorch, OpenCV, NumPy* Link | Apr 2026

- Designed a multimodal ensemble that routes faces, video frames, and audio to four different models (EfficientNet, MesoNet, XceptionNet, AASIST) to detect general & deepfake AI-generated video.
- Fine-tuned a pre-trained EfficientNet-B1 model on datasets such as AIGVDBench and FaceForensics++, which use diffusion and face transfer methods to generate deepfake videos.
- Integrated the EfficientNet-B1 inference logic into the ensemble, utilizing MTCNN for face detection and cropping, and Albumentations for high-performance preprocessing.
- Designed an ensemble architectural diagram including preprocessing, inference, and postprocessing stages.
- Built the ensemble's parallel inference pipeline, fusing model scores by mean averaging or stacking (five fusion strategies are compared in the paper) into the final REAL/FAKE prediction; communicated backend results with the frontend through HTTP requests.
- Analyzed the ensemble's ∼70% accuracy on the unseen FakeAVCeleb dataset, against 90%+ in-domain accuracy for individual models such as XceptionNet, and attributed the gap mainly to domain shift.
- Recruited three fellow undergraduates and agreed on the project scope with teammates and the supervisor; served as a senior developer on the 5-member team, participating in two weekly meetings — a Sunday sync facilitated by the mentor to review weekly progress, identify blockers, and plan upcoming tasks, and a Monday meeting to present results to the professor.

Sycophancy Bias Replication (TAISI Intensive) | *Python, OpenRouter API* Link | Jul 2026

- Replicated a 2024 research finding on sycophancy bias in LLMs, testing whether newer models resist "suggested-answer" bias where a user states a belief about the answer before asking a factual question.
- Built an evaluation pipeline against the OpenRouter API to benchmark Claude, GPT, and Gemini models on 60 sampled multiple-choice questions from MMLU, TruthfulQA, LogiQA, and HellaSwag, under biased and unbiased conditions.
- Computed a "biasing rate" metric per model to quantify how often each model switched its answer to match an incorrectly suggested option, comparing robustness across model generations.

JetBrains LCA Benchmark Contribution | *Python, Docker, OpenAI API, Hugging Face API* Link | Apr 2026

- Contributed to the open-source JetBrains LCA benchmark, expanding the bug localization benchmark's capability from identifying the buggy file to automatically applying a fix.
- Used the OpenAI API to generate code patches and the Hugging Face API to programmatically download and manage large-scale repositories.
- Expanded evaluation metrics of the original benchmark by adding token usage count, iteration count, etc. for popular coding agents like Claude Code and Codex.
- Co-authored a technical white paper detailing the multi-agent architecture and evaluation metrics used to benchmark Claude Code and Codex; delivered a group presentation to **40+ attendees**.

Third/Fourth Year Courses

- **AI & Robotics:** Introduction to Machine Learning, Introduction to Artificial Intelligence, Fundamentals of Robotics
- **Systems & Networks:** Operating Systems, Parallel Programming, Principles of Computer Networks
- **Algorithms:** Algorithm Design and Analysis
- **In Progress:** Neural Networks and Deep Learning (CSC413), Robotic Perception (CSC478), Capstone Design Course (CSC490)